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Optical lens assembly and photographing module

Abstract

An optical lens assembly includes, in order from an object side to an image side: a first lens with negative refractive power; a second lens with positive refractive power; a third lens with positive refractive power; a fourth lens with positive refractive power; a fifth lens with negative refractive power; a sixth lens with positive refractive power; wherein a chief ray angle incident on an image plane at a maximum view angle of the optical lens assembly is CRA, a distance from the image-side surface of the sixth lens to the image plane along an optical axis is BFL, a focal length of the optical lens assembly is f, and the following condition is satisfied: $0.54° / mm^2 < CRA / * BFL * f) < 17.33 / mm^2$.

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Background/Summary

BACKGROUND
Field of the Invention
(1) The present invention relates to an optical lens assembly and a photographing module, and more particularly to an optical lens assembly and a photographing module applicable to electronic products.
Description of Related Art
(2) Small camera lens devices can be widely used in various electronic devices, such as, wearable display, smart phone, tablet computer, game player, dashcam, household electronic device, surveillance camera and so on. With the emergence of Internet of Things (IoT), the demand for the Internet of Things devices is also increasing Therefore, how to develop a small camera lens device to have a large field of view, be workable in both the visible and infrared light bands, and adapt to a large temperature range while being applied to Internet of Things devices is the motivation of the present invention.
(3) The present invention mitigates and/or obviates the aforementioned disadvantages.
SUMMARY

(4) The objective of the present invention is to provide an optical lens assembly and a photographing module, and the optical lens assembly has a total of six lenses with refractive power. When a specific condition is satisfied, the optical lens assembly has the image quality with high image resolution in both visible and infrared light bands, and can achieve ultra-wide field of view and reduce the influence of ambient temperature on image quality.
(5) In addition, when the lens is made of glass, the optical lens assembly of the present invention can be used at more extreme temperatures.
(6) Therefore, an optical lens assembly in accordance with the present invention includes, in order from an object side to an image side: a first lens with negative refractive power, including an object-side surface and an image-side surface, the object-side surface of the first lens being concave in a paraxial region thereof, and the image-side surface of the first lens being concave in a paraxial region thereof; a second lens with positive refractive power, including an object-side surface and an image-side surface, the object-side surface of the second lens being concave in a paraxial region thereof, and the image-side surface of the second lens being convex in a paraxial region thereof; a third lens with positive refractive power, including an object-side surface and an image-side surface, the object-side surface of the third lens being concave in a paraxial region thereof, and the image-side surface of the third lens being convex in a paraxial region thereof; a fourth lens with positive refractive power, including an object-side surface and an image-side surface, the object-side surface of the fourth lens being convex in a paraxial region thereof, and the image-side surface of the fourth lens being convex in a paraxial region thereof; a fifth lens with negative refractive power, including an object-side surface and an image-side surface, the object-side surface of the fifth lens being concave in a paraxial region thereof, and the image-side surface of the fifth lens being concave in a paraxial region thereof; and a sixth lens with positive refractive power, including an object-side surface and an image-side surface, and the object-side surface of the sixth lens being convex in a paraxial region thereof.
(7) Wherein a chief ray angle incident on an image plane at a maximum view angle of the optical lens assembly is CRA, a distance from the image-side surface of the sixth lens to the image plane along the optical axis is BFL, a focal length of the optical lens assembly is f, and the following condition is satisfied:
(8) $0.54° / mm^2 < CRA / * BFL * f) < 17.33 / mm^2$.
(9) The optical lens assembly has a total of six lenses with refractive power.
(10) The present invention has the following effect. When 0.54°/mm.sup.2<CRA/(BFL*f)<17.33°/mm.sup.2 is satisfied, the incident angle of the chief ray, the rear focal space and the ratio of the focal length of the system are more appropriate, which can provide better relative illumination and meet the rear focal requirements of the module. Optionally, the following condition is satisfied: 8.85°/mm.sup.2<CRA/(BFL*f)<13.88°/mm.sup.2. When the third lens is made of glass, the optical lens assembly of the present invention can be used at more extreme temperatures.
(11) A radius of curvature of the object-side surface of the third lens is R5, a radius of curvature of the image-side surface of the third lens is R6, and the following condition is satisfied: 1.07<R5/R6<3.56, so that the ratio of the curvatures of the third lenses is better to reduce the difficulty in manufacturing glass lenses. Optionally, the following condition is satisfied:
(12) $1.48 < R5 / R6 < 2.63$.
(13) The radius of curvature of the object-side surface of the third lens is R5, the radius of curvature of the image-side surface of the third lens is R6, a focal length of the third lens is f3, a thickness of the third lens along the optical axis is CT3, and the following condition is satisfied: 0.88<(R5*R6)/(f3*CT3)<11.25, so that the proportion of glass lens parameters of the third lens is better to improve the glass manufacturability and reduce the temperature drift problem of the optical lens assembly. Optionally, the following condition is satisfied.
(14) $2.24 < (R5 * R6) / (f3 * CT3) < 9.17$.

(15) A focal length of the sixth lens is f6, a radius of curvature of the object-side surface of the sixth lens is R11, and the following condition is satisfied: 0.89<f6/R11<4.35, so that the curvature and refractive power design of the sixth lens are better to improve the image quality of the optical lens assembly and reduce the lens sensitivity. Optionally, the following condition is satisfied:
(16) $1.68 < f6 / R11 < 4.35$.
(17) A distance from the object-side surface of the first lens to the image plane along the optical axis is TL, the distance from the image-side surface of the sixth lens to the image plane along the optical axis is BFL, a maximum image height of the optical lens assembly is IMH, and the following condition is satisfied: 3.74<(TL-BFL)/IMH<6.84, so that the module height, the rear focal space and the aspect ratio of the image are better to achieve the optimal module height and imaging size requirements. Optionally, the following condition is satisfied:
(18) $4.13 < (TL - BFL) / IMH < 6.34$.
(19) A maximum field of view of the optical lens assembly is FOV, the chief ray angle incident on the image plane at the maximum view angle of the optical lens assembly is CRA, and the following condition is satisfied: 4.23<FOV/CRA<13.32, so that the cooperation of the incident angle of the chief ray and the ultra-wide field of view characteristic is better to reduce the aberration of the optical lens assembly. Optionally, the following condition is satisfied:
(20) $4.23 < FOV / CRA < 6.36$.
(21) A radius of curvature of the image-side surface of the fifth lens is R10, the radius of curvature of the object-side surface of the sixth lens is R11, and the following condition is satisfied: 0.67<R10/R11<7.20, so that the proportion of refractive power of the lenses is better to reduce the ghost problem between the lenses. Optionally, the following condition is satisfied:
(22) $0.98 < R10 / R11 < 7.2$.
(23) The chief ray angle incident on the image plane at the maximum view angle of the optical lens assembly is CRA, the distance from the image-side surface of the sixth lens to the image plane along the optical axis is BFL, the maximum image height of the optical lens assembly is IMH, and the following condition is satisfied: 0.27<tan(CRA)*BFL/IMH<0.47, so that the proportion of the optical lens assembly is better to meet the rear focal space and image sensor imaging size and the incident angle requirements. Optionally, the following condition is satisfied:
(24) $0.29 < \tan(CRA) * BFL / IMH < 0.46$.
(25) An Abbe number of the third lens is vd3, a refractive index of the third lens is nd3, the focal length of the third lens is f3, the thickness of the third lens along the optical axis is CT3, and the following condition is satisfied: 0.87 mm.sup.−2<vd3/(nd3*f3*CT3)<5.74 mm.sup.−2, so that the lens material, refractive power and thickness design ratio of the third lens is better to provide better glass lens manufacturability. Optionally, the following condition is satisfied:
(26) $01.61mm^{-2} < vd3 / (nd3 * f3 * CT3) < 4.02mm^{-2}$.
(27) The focal length of the optical lens assembly is f, a focal length of the first lens is f1, and the following condition is satisfied: −0.68<f/f1<−0.34, so that the ratio of the focal length of the first lens to that of the optical lens assembly can enhance the wide-field of view characteristic of the optical lens assembly, so as to provide a larger field of view. Optionally, the following condition is satisfied:
(28) $0.58 < f / f1 < -0.38$.
(29) A focal length of the fourth lens is f4, a focal length of the fifth lens is f5, and the following condition is satisfied: −2.24<f4/f5<−**0.77**, so that the distribution of the refractive power of the optical lens assembly is be more appropriate, which is favorable to correcting the aberration of the optical lens assembly for the enhancement of the image quality. Optionally, the following condition is satisfied:
(30) $-1.59 < f4 / f5 < -0.77$.
(31) The focal length of the fourth lens is f4, the focal length of the fifth lens is f5, a radius of curvature of the image-side surface of the fourth lens is R8, a radius of curvature of the object-side

surface of the fifth lens is R9, and the following condition is satisfied: −1.71<f4*R8/(R9*f5) <−0.66, so that the curvature and refractive power design of the optical lens assembly are better to improve the confocal properties of the visible and infrared light bands. Optionally, the following condition is satisfied:
(32) $-1.35 < f4 * R8 / (R9 * f5) < -0.73$.
(33) The distance from the object-side surface of the first lens to the image plane along the optical axis is TL, the maximum image height of the optical lens assembly is IMH, the focal length of the optical lens assembly is f, and the following condition is satisfied: 1.59 mm.sup.−1<TL/(IMH*f) <5.84 mm.sup.−1, so that the module height, the image height and the refractive power ratio of the optical lens assembly are better to achieve the optimum imaging effects. Optionally, the following condition is satisfied:
(34) $2.94mm^{-1} < TL / (IMH * f) < 4.47mm^{-1}$.
(35) A focal length of the second lens is f2, the focal length of the third lens is f3, and the following condition is satisfied: 0.74<f2/f3<4.70, so that the proportion of the refractive power ratio of the lenses of the optical lens assembly is better to reduce the temperature drift problem. Optionally, the following condition is satisfied:
(36) $1.04 < f2 / f3 < 3.25$.
(37) The radius of curvature of the image-side surface of the fourth lens is R8, the radius of curvature of the object-side surface of the fifth lens is R9, a distance from the fourth lens to the fifth lens along the optical axis is T45, and the following condition is satisfied: 2.21 mm.sup. −1<(R8/R9)/T45<15.14 mm.sup.−1, so that the proportion of the refractive power and interval of the lenses of the optical lens assembly is better to reduce the ghost problem between the lenses. Optionally, the following condition is satisfied:
(38) $6.15mm^{-1} < (R8 / R9) / T45 < 15.14mm^{-1}$.
(39) A thickness of the fourth lens along the optical axis is CT4, a thickness of the fifth lens along the optical axis is CT5, a thickness of the sixth lens along the optical axis is CT6, the focal length of the fifth lens is f5, and the following condition is satisfied: −3.28 mm.sup. −1<(CT4+CT6)/(CT5*f5)<−0.71 mm.sup.−1, so that the thicknesses and refractive powers of the lenses are better to enhance the image quality and reduce the lens tolerance. Optionally, the following condition is satisfied:
(40) $-1.23mm^{-1} < (CT4 + CT6) / (CT5 * f5) < -0.71mm^{-1}$.
(41) The focal length of the fifth lens is f5, the focal length of the sixth lens is f6, and the following condition is satisfied: −12.96 mm<f5*f6/f<−5.07 mm, so that the distribution of the refractive power of the lenses of the optical lens assembly is better, and the optical lens assembly can be applied in both the visible and infrared light bands.
(42) Moreover, a photographing module in accordance with the present invention includes a lens barrel, the aforementioned optical lens assembly disposed in the lens barrel, and an image sensor disposed on the image plane of the optical lens assembly.
(43) The present invention will be presented in further details from the following descriptions with the accompanying drawings, which show, for purpose of illustrations only, the preferred embodiments in accordance with the present invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**A is a schematic view of an optical lens assembly in accordance with a first embodiment of the present invention;
(2) FIG. **1**B is a schematic diagram showing, in order from left to right, the field curvature curve and the distortion curve of the first embodiment of the present invention, which illustrate the states

of the field curvature curve and the distortion curve in both visible and infrared light bands;
(3) FIG. **1**C is a schematic view of parameter CRA in accordance with a first embodiment of the present invention;
(4) FIG. **2**A is a schematic view of an optical lens assembly in accordance with a second embodiment of the present invention;
(5) FIG. **2**B is a schematic diagram showing, in order from left to right, the field curvature curve and the distortion curve of the second embodiment of the present invention, which illustrate the states of the field curvature curve and the distortion curve in both visible and infrared light bands;
(6) FIG. **3**A is a schematic view of an optical lens assembly in accordance with a third embodiment of the present invention;
(7) FIG. **3**B is a schematic diagram showing, in order from left to right, the field curvature curve and the distortion curve of the third embodiment of the present invention, which illustrate the states of the field curvature curve and the distortion curve in both visible and infrared light bands;
(8) FIG. **4**A is a schematic view of an optical lens assembly in accordance with a fourth embodiment of the present invention;
(9) FIG. **4**B is a schematic diagram showing, in order from left to right, the field curvature curve and the distortion curve of the fourth embodiment of the present invention, which illustrate the states of the field curvature curve and the distortion curve in both visible and infrared light bands;
(10) FIG. **5**A is a schematic view of an optical lens assembly in accordance with a fifth embodiment of the present invention;
(11) FIG. **5**B is a schematic diagram showing, in order from left to right, the field curvature curve and the distortion curve of the fifth embodiment of the present invention, which illustrate the states of the field curvature curve and the distortion curve in both visible and infrared light bands;
(12) FIG. **6**A is a schematic view of an optical lens assembly in accordance with a sixth embodiment of the present invention;
(13) FIG. **6**B is a schematic diagram showing, in order from left to right, the field curvature curve and the distortion curve of the sixth embodiment of the present invention, which illustrate the states of the field curvature curve and the distortion curve in both visible and infrared light bands; and
(14) FIG. **7** is a schematic view of a photographing module in accordance with a seventh embodiment of the present invention.

DETAILED DESCRIPTION

First Embodiment

(15) Referring to FIGS. **1**A, **1**B and **1**C, FIG. **1**A shows a schematic view of an optical lens assembly in accordance with a first embodiment of the present invention, FIG. **1**B shows, in order from left to right, the field curvature curve and the distortion curve of the first embodiment of the present invention, which illustrate the states of the field curvature curve and the distortion curve in both visible and infrared light bands, and FIG. **1**C is a schematic view of parameter CRA in accordance with a first embodiment of the present invention. As shown in FIG. **1**A, the optical lens assembly includes, in order from an object side to an image side along an optical axis **190**: a first lens **110**, a second lens **120**, a stop **100**, a third lens **130**, a fourth lens **140**, a fifth lens **150**, a sixth lens **160**, an optical filter **170**, and an image plane **182**. The optical lens assembly can cooperate with an image sensor **181** disposed on an image plane **182**. The optical lens assembly has a total of six lenses with refractive power, but not is limited thereto.
(16) The first lens **110** with negative refractive power includes an object-side surface **111** and an image-side surface **112**, the object-side surface **111** of the first lens **110** is concave in a paraxial region thereof, the image-side surface **112** of the first lens **110** is concave in a paraxial region thereof, the object-side surface **111** and the image-side surface **112** of the first lens **110** are aspheric, and the first lens **110** is made of plastic.
(17) The second lens **120** with positive refractive power includes an object-side surface **121** and an image-side surface **122**, the object-side surface **121** of the second lens **120** is concave in a paraxial

region thereof, the image-side surface **122** of the second lens **120** is convex in a paraxial region thereof, the object-side surface **121** and the image-side surface **122** of the second lens **120** are aspheric, and the second lens **120** is made of plastic.
(18) The third lens **130** with positive refractive power includes an object-side surface **131** and an image-side surface **132**, the object-side surface **131** of the third lens **130** is concave in a paraxial region thereof, the image-side surface **132** of the third lens **130** is convex in a paraxial region thereof, the object-side surface **131** and the image-side surface **132** of the third lens **130** are aspheric, and the third lens **130** is made of glass.
(19) The fourth lens **140** with positive refractive power includes an object-side surface **141** and an image-side surface **142**, the object-side surface **141** of the fourth lens **140** is convex in a paraxial region thereof, the image-side surface **142** of the fourth lens **140** is convex in a paraxial region thereof, the object-side surface **141** and the image-side surface **142** of the fourth lens **140** are aspheric, and the fourth lens **140** is made of plastic.
(20) The fifth lens **150** with negative refractive power includes an object-side surface **151** and an image-side surface **152**, the object-side surface **151** of the fifth lens **150** is concave in a paraxial region thereof, the image-side surface **152** of the fifth lens **150** is concave in a paraxial region thereof, the object-side surface **151** and the image-side surface **152** of the fifth lens **150** are aspheric, and the fifth lens **150** is made of plastic.
(21) The sixth lens **160** with positive refractive power includes an object-side surface **161** and an image-side surface **162**, the object-side surface **161** of the sixth lens **160** is convex in a paraxial region thereof, the image-side surface **162** of the sixth lens **160** is concave in a paraxial region thereof, the object-side surface **161** and the image-side surface **162** of the sixth lens **160** are aspheric, and the sixth lens **160** is made of plastic.
(22) The optical filter **170** is made of glass, is located between the sixth lens **160** and the image plane **182**, and has no influence on the focal length of the optical lens assembly. In the present embodiment, the optical filter **170** is selected from filters that allow light in the visible light wavelengths, in the infrared light wavelengths or in both the visible and infrared light wavelengths to pass therethrough.
(23) The curve equation for the aspheric surface profiles of the respective lenses of the first embodiment is expressed as follows:
(24) $z(h) = \frac{ch^2}{1 + [1 - (k+1)c^2 h^2]^{0.5}} + \text{ .Math. } (A_i) \text{ .Math. } (h^i)$
wherein: z represents the value of a reference position at a height of h with respect to a vertex of the surface of a lens along the optical axis **190**; c represents a paraxial curvature (i.e., a curvature of a lens surface in a paraxial region thereof) equal to 1/R (R: a paraxial radius of curvature); h represents a vertical distance from the point on the curve of the aspheric surface to the optical axis **190**; k represents the conic constant; and Ai represents the i-th order aspheric coefficient.
(25) In the first embodiment of the optical lens assembly, a focal length of the optical lens assembly is f, a f-number of the optical lens assembly is Fno, a maximum field of view of the optical lens assembly is FOV, an entrance pupil diameter of the optical lens assembly is EPD, and their values are expressed as follows: f=1.60 mm; Fno=2.20; FOV=151.0 degrees; and EPD=0.71 mm.
(26) In the first embodiment of the optical lens assembly, a chief ray **14** angle incident on the image plane **182** at a maximum view angle of the optical lens assembly is CRA (as shown in FIG. **1**C), a distance from the image-side surface **162** of the sixth lens **160** to the image plane **182** along the optical axis **190** is BFL, the focal length of the optical lens assembly is f, and the following condition is satisfied:
(27) $CRA / (BFL * f) = 11.39° / mm^2$.
(28) In the first embodiment of the optical lens assembly, a radius of curvature of the object-side surface **131** of the third lens **130** is R5, a radius of curvature of the image-side surface **132** of the third lens **130** is R6, and the following condition is satisfied:

(29) $0R5 / R6 = 2.19$.
(30) In the first embodiment of the optical lens assembly, the radius of curvature of the object-side surface **131** of the third lens **130** is R5, the radius of curvature of the image-side surface **132** of the third lens **130** is R6, a focal length of the third lens **130** is f3, a central thickness of the third lens **130** along the optical axis **190** is CT3, and the following condition is satisfied:
(31) $(R5 * R6) / (f3 * \mathrm{CT3}) = 5.16$.
(32) In the first embodiment of the optical lens assembly, a focal length of the sixth lens **160** is f6, a radius of curvature of the object-side surface **161** of the sixth lens **160** is R11, and the following condition is satisfied:
(33) $f6 / R11 = 2.84$.
(34) In the first embodiment of the optical lens assembly, a distance from the object-side surface **111** of the first lens **110** to the image plane **182** along the optical axis **190** is TL, the distance from the image-side surface **162** of the sixth lens **160** to the image plane **182** along the optical axis **190** is BFL, a maximum image height of the optical lens assembly is IMH, and the following condition is satisfied:
(35) $(\mathrm{TL} - \mathrm{BFL}) / \mathrm{IMH} = 5.27$.
(36) In the first embodiment of the optical lens assembly, the maximum field of view of the optical lens assembly is FOV, the chief ray **14** angle incident on the image plane **182** at the maximum view angle of the optical lens assembly is CRA, and the following condition is satisfied:
(37) $\mathrm{FOV} / \mathrm{CRA} = 5.28$.
(38) In the first embodiment of the optical lens assembly, a radius of curvature of the image-side surface **152** of the fifth lens **150** is R10, the radius of curvature of the object-side surface **161** of the sixth lens **160** is R11, and the following condition is satisfied:
(39) $R10 / R11 = 1.64$.
(40) In the first embodiment of the optical lens assembly, the chief ray **14** angle incident on the image plane **182** at the maximum view angle of the optical lens assembly is CRA, the distance from the image-side surface **162** of the sixth lens **160** to the image plane **182** along the optical axis **190** is BFL, the maximum image height of the optical lens assembly is IMH, and the following condition is satisfied:
(41) $\tan(\mathrm{CRA}) * \mathrm{BFL} / \mathrm{IMH} = 0.37$.
(42) In the first embodiment of the optical lens assembly, an Abbe number of the third lens **130** is vd3, a refractive index of the third lens is nd3, the focal length of the third lens **130** is f3, the central thickness of the third lens **130** along the optical axis **190** is CT3, and the following condition is satisfied:
(43) $\mathrm{vd3} / (\mathrm{nd3} * f3 * \mathrm{CT3}) = 2.98\mathrm{mm}^{-2}$.
(44) In the first embodiment of the optical lens assembly, the focal length of the optical lens assembly is f, a focal length of the first lens **110** is f1, and the following condition is satisfied:
(45) $f / f1 = -0.48$.
(46) In the first embodiment of the optical lens assembly, a focal length of the fourth lens **140** is f4, a focal length of the fifth lens **150** is f5, and the following condition is satisfied:
(47) $f4 / f5 = -1.14$.
(48) In the first embodiment of the optical lens assembly, the focal length of the fourth lens **140** is f4, the focal length of the fifth lens **150** is f5, a radius of curvature of the image-side surface **142** of the fourth lens **140** is R8, a radius of curvature of an object-side surface **151** of the fifth lens **150** is R9, and the following condition is satisfied:
(49) $0f4 * R8 / (R9 * f5) = -0.92$.
(50) In the first embodiment of the optical lens assembly, the distance from the object-side surface **111** of the first lens **110** to the image plane **182** along the optical axis **190** is TL, the maximum image height of the optical lens assembly is IMH, the focal length of the optical lens assembly is f, and the following condition is satisfied:

(51) $TL / (IMH * f) = 3.72mm^{-1}$.
(52) In the first embodiment of the optical lens assembly, the focal length of the second lens **120** is f2, the focal length of the third lens **130** is f3, and the following condition is satisfied:
(53) $f2 / f3 = 2.08$.
(54) In the first embodiment of the optical lens assembly, the radius of curvature of the image-side surface **142** of the fourth lens **140** is R8, the radius of curvature of an object-side surface **151** of the fifth lens **150** is R9, a distance from the fourth lens **140** to the fifth lens **150** along the optical axis **190** is T45, and the following condition is satisfied:
(55) $(R8 / R9) / T45 = 12.62mm^{-1}$.
(56) In the first embodiment of the optical lens assembly, a central thickness of the fourth lens **140** along the optical axis **190** is CT4, a central thickness of the fifth lens **150** along the optical axis **190** is CT5, a central thickness of the sixth lens **160** along the optical axis **190** is CT6, the focal length of the fifth lens **150** is f5, and the following condition is satisfied:
(57) $(CT4 + CT6) / (CT5 * f5) = -0.94mm^{-1}$.
(58) In the first embodiment of the optical lens assembly, the focal length of the fifth lens **150** is f5, the focal length of the sixth lens **160** is f6, the focal length of the optical lens assembly is f, and the following condition is satisfied:
(59) $f5 * f6 / f = -9.02mm$.
(60) The detailed optical data of the respective elements in the optical lens assembly of the first embodiment is shown in Table 1, and the aspheric coefficients of the lenses in the first embodiment is shown in Table 2.
(61) TABLE-US-00001 TABLE 1 Embodiment 1 f = 1.60 mm, Fno = 2.20, FOV = 151.0° Radius Refractive Abbe of Thickness/ index number Focal Surface curvature gap Material (nd) (vd) length 0 Object Infinity Infinity 1 First lens −18.032 (ASP) 1.236 plastic 1.544 56.0 −3.33 2 2.068 (ASP) 1.978 3 Second lens −47.658 (ASP) 3.447 plastic 1.661 20.4 18.51 4 −10.084 (ASP) 0.636 5 Stop Infinity 0.070 6 Third lens −9.263 (ASP) 0.854 glass 1.806 40.7 8.89 7 −4.231 (ASP) 0.035 8 Fourth lens 2.821 (ASP) 1.443 plastic 1.544 56.0 2.54 9 −2.228 (ASP) 0.064 10 Fifth lens −2.777 (ASP) 1.208 plastic 1.661 20.4 −2.22 11 3.731 (ASP) 0.114 12 Sixth lens 2.280 (ASP) 1.077 plastic 1.544 56.0 6.48 13 5.347 (ASP) 0.547 14 Optical filter Infinity 0.210 glass 1.517 64.2 15 Infinity 0.814 16 Image plane Infinity —
(62) TABLE-US-00002 TABLE 2 Aspheric Coefficients Surface 1 2 3 4 6 7 K: 6.4800E+00 −8.6507E−01 −9.9802E+01 5.0719E+01 −2.1938E+00 1.1436E+00 A2: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A4: 3.8082E−03 1.6649E−03 −1.7407E−02 5.7815E−03 5.3062E−03 2.7514E−03 A6: −1.6758E−04 5.5762E−04 1.6747E−03 3.8495E−03 −6.3183E−03 −8.0292E−03 A8: 5.6201E−06 5.9747E−04 −4.0844E−04 −9.5378E−04 9.8648E−04 −2.1954E−03 A10: −1.0583E−07 −1.5350E−04 8.0077E−06 6.0402E−04 9.8648E−04 8.8646E−04 A12: 9.4723E−10 2.5800E−05 2.0196E−05 −5.7793E−04 0.0000E+00 0.0000E+00 A14: 7.3555E−14 −1.9800E−06 −2.3523E−06 3.6401E−04 0.0000E+00 0.0000E+00 A16: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A18: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A20: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 Surface 8 9 10 11 12 13 K −1.4538E+00 2.6202E−01 −1.6745E+00 1.3510E+00 −4.2405E+00 −1.7290E+01 A2: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A4: −8.1101E−03 −2.0037E−02 −9.3177E−03 −2.2117E−03 −2.6424E−02 −1.1740E−02 A6: −3.9557E−03 1.2292E−02 −8.4655E−03 2.5953E−03 1.1146E−02 −2.8279E−03 A8: −3.6503E−03 3.4465E−05 5.7633E−03 2.6178E−04 −1.1823E−03 2.0901E−03 A10: −2.2269E−05 −4.4559E−04 1.1651E−03 −5.6003E−04 1.0273E−04 −2.3324E−04 A12: 6.4766E−04 1.8465E−05 −2.0152E−04 2.0305E−04 −6.5631E−06 −3.9733E−05 A14: −3.6092E−06 1.3402E−04 −9.6461E−05 −3.7017E−05 −7.4755E−07 1.3103E−05 A16: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A18: 0.0000E+00

0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A20: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00

(63) In Table 1, the units of the radius of curvature, the thickness, the gap and the focal length are expressed in mm, and the surface numbers **0-16** respectively represent the surfaces from the object-side to the image-side, wherein the surface **0** represents a gap between an object and the object-side surface **111** of the first lens **110** along the optical axis **190**; the surface **1** represents the thickness of the first lens **110** along the optical axis **190**; the surface **3** represents the thickness of the second lens **120** along the optical axis **190**; the surface **6** represents the thickness of the third lens **130** along the optical axis **190**; the surface **8** represents the thickness of the fourth lens **140** along the optical axis **190**; the surface **10** represents the thickness of the fifth lens **150** along the optical axis **190**; the surface **12** represents the thickness of the sixth lens **160** along the optical axis **190**; the surface **14** represents the thickness of the optical filter **170** along the optical axis **190**; the surface **2** represents a gap between the first lens **110** and the second lens **120** along the optical axis **190**; the surface **4** represents a gap between the second lens **120** and the stop **100** along the optical axis **190**; the surface **5** represents a gap between the stop **100** and the third lens **130** along the optical axis **190**, which is expressed as a positive value since the stop **100** is closer to the object-side than the object-side surface **131** of the third lens **130**; the surface **7** represents a gap between the third lens **130** and the fourth lens **140** along the optical axis **190**; the surface **9** represents a gap between the fourth lens **140** and the fifth lens **150** along the optical axis **190**; the surface **11** represents a gap between the fifth lens **150** and the sixth lens **160** along the optical axis **190**; the surface **13** represents a gap between the sixth lens **160** and the optical filter **170** along the optical axis **190**; and the surface **15** represents a gap between the optical filter **170** and the image plane **182** along the optical axis **190**.

(64) In table 2, k represents the conic constant of the equation of aspheric surface profiles, and A2, A4, A6, A8, A10, A12, A14, A16, A18, and A20 represent the high-order aspheric coefficients. The respective tables presented below for respective one of other embodiments are based on the schematic view of this embodiment, and the definitions of parameters in the tables are the same as those in Tables 1-2 of the first embodiment. Therefore, an explanation in this regard will not be provided again.

Second Embodiment

(65) Referring to FIGS. **2**A and **2**B, FIG. **2**A shows a schematic view of an optical lens assembly in accordance with a second embodiment of the present invention, FIG. **2**B shows, in order from left to right, the field curvature curve and the distortion curve of the second embodiment of the present invention, which illustrate the states of the field curvature curve and the distortion curve in both visible and infrared light bands. As shown in FIG. **2**A, the optical lens assembly includes, in order from an object side to an image side along an optical axis **290**: a first lens **210**, a second lens **220**, a stop **200**, a third lens **230**, a fourth lens **240**, a fifth lens **250**, a sixth lens **260**, an optical filter **270**, and an image plane **282**. The optical lens assembly can cooperate with an image sensor **281** disposed on an image plane **282**. The optical lens assembly has a total of six lenses with refractive power, but not is limited thereto.

(66) The first lens **210** with negative refractive power includes an object-side surface **211** and an image-side surface **212**, the object-side surface **211** of the first lens **210** is concave in a paraxial region thereof, the image-side surface **212** of the first lens **210** is concave in a paraxial region thereof, the object-side surface **211** and the image-side surface **212** of the first lens **210** are aspheric, and the first lens **210** is made of plastic.

(67) The second lens **220** with positive refractive power includes an object-side surface **221** and an image-side surface **222**, the object-side surface **221** of the second lens **220** is concave in a paraxial region thereof, the image-side surface **222** of the second lens **220** is convex in a paraxial region thereof, the object-side surface **221** and the image-side surface **222** of the second lens **220** are aspheric, and the second lens **220** is made of plastic.

(68) The third lens **230** with positive refractive power includes an object-side surface **231** and an image-side surface **232**, the object-side surface **231** of the third lens **230** is concave in a paraxial region thereof, the image-side surface **232** of the third lens **230** is convex in a paraxial region thereof, the object-side surface **231** and the image-side surface **232** of the third lens **230** are aspheric, and the third lens **230** is made of glass.
(69) The fourth lens **240** with positive refractive power includes an object-side surface **241** and an image-side surface **242**, the object-side surface **241** of the fourth lens **240** is convex in a paraxial region thereof, the image-side surface **242** of the fourth lens **240** is convex in a paraxial region thereof, the object-side surface **241** and the image-side surface **242** of the fourth lens **240** are aspheric, and the fourth lens **240** is made of plastic.
(70) The fifth lens **250** with negative refractive power includes an object-side surface **251** and an image-side surface **252**, the object-side surface **251** of the fifth lens **250** is concave in a paraxial region thereof, the image-side surface **252** of the fifth lens **250** is concave in a paraxial region thereof, the object-side surface **251** and the image-side surface **252** of the fifth lens **250** are aspheric, and the fifth lens **250** is made of plastic.
(71) The sixth lens **260** with positive refractive power includes an object-side surface **261** and an image-side surface **262**, the object-side surface **261** of the sixth lens **260** is convex in a paraxial region thereof, the image-side surface **262** of the sixth lens **260** is concave in a paraxial region thereof, the object-side surface **261** and the image-side surface **262** of the sixth lens **260** are aspheric, and the sixth lens **260** is made of plastic.
(72) The optical filter **270** is made of glass, is located between the sixth lens **260** and the image plane **282**, and has no influence on the focal length of the optical lens assembly. In the present embodiment, the optical filter **270** is selected from filters that allow light in the visible light wavelengths, in the infrared light wavelengths or in both the visible and infrared light wavelengths to pass therethrough.
(73) The detailed optical data of the respective elements in the optical lens assembly of the second embodiment is shown in Table 3, and the aspheric coefficients of the lenses in the second embodiment is shown in Table 4.
(74) TABLE-US-00003 TABLE 3 Embodiment 2 f = 1.60 mm, Fno = 2.20, FOV = 151.0° Radius Refractive Abbe of Thickness/ index number Focal Surface curvature gap Material (nd) (vd) length 0 Object Infinity Infinity 1 First lens −19.773 (ASP) 1.537 plastic 1.544 56.0 −3.31 2 2.041 (ASP) 1.891 3 Second lens −37.686 (ASP) 3.032 plastic 1.661 20.4 17.95 4 −9.372 (ASP) 0.809 5 Stop Infinity 0.074 6 Third lens −13.620 (ASP) 0.815 glass 1.806 40.7 13.76 7 −6.293 (ASP) 0.040 8 Fourth lens 2.168 (ASP) 1.520 plastic 1.544 56.0 2.25 9 −2.129 (ASP) 0.103 10 Fifth lens −1.838 (ASP) 1.172 plastic 1.661 20.4 −2.32 11 12.301 (ASP) 0.042 12 Seventh lens 2.050 (ASP) 0.888 plastic 1.544 56.0 7.43 13 3.512 (ASP) 0.526 14 Optical filter Infinity 0.210 glass 1.517 64.2 15 Infinity 0.877 16 Image plane Infinity —
(75) TABLE-US-00004 TABLE 4 Aspheric Coefficients Surface 1 2 3 4 6 7 K: 7.3631E+00 −8.5693E−01 2.6441E+01 1.5740E+01 0.0000E+00 0.0000E+00 A2: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A4: 3.4251E−03 4.6821E−03 −1.4467E−02 7.9229E−03 3.2489E−02 1.4854E−03 A6: −1.5430E−04 1.5062E−03 1.7731E−03 −2.9248E−03 −2.0378E−02 −1.0157E−02 A8: 5.2803E−06 2.3959E−04 −7.7514E−04 6.7067E−05 0.0000E+00 0.0000E+00 A10: −1.0809E−07 −3.4239E−05 4.3488E−05 4.3231E−04 0.0000E+00 0.0000E+00 A12: 1.2384E−09 7.2912E−06 2.9361E−05 −2.0180E−04 0.0000E+00 0.0000E+00 A14: −5.0892E−12 −1.2228E−06 −3.6805E−06 2.3532E−05 0.0000E+00 0.0000E+00 A16: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A18: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A20: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 Surface 8 9 10 11 12 13 K: −2.6116E+00 6.7630E−02 −4.2512E+00 2.8260E+01 −4.9619E+00 −5.7951E+00 A2 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A4: −1.4270E−02

−1.6687E−02 −1.2574E−02 3.5761E−02 −2.1072E−02 −3.0853E−02 A6: −9.7908E−04 1.4355E−02 −8.4726E−03 6.7284E−05 7.5046E−03 3.4841E−03 A8: −1.3066E−03 −2.0293E−03 9.5779E−03 −3.9852E−04 −6.6367E−04 1.1506E−03 A10: −1.8433E−03 1.0247E−03 −1.4932E−04 −2.9977E−04 2.0516E−04 −4.1726E−04 A12: 1.1264E−03 −2.1187E−04 −7.4822E−04 1.2385E−04 −3.6165E−05 1.1857E−05 A14: −9.1470E−05 4.7940E−05 1.4920E−04 −4.7060E−05 −2.2317E−06 2.2161E−05 A16: 0.0000E+00 0.0000E+00 −2.6438E−05 5.4942E−06 5.5794E−07 −2.8203E−06 A18: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A20: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00

(76) In the second embodiment, the curve equation of the aspheric surface profiles of the aforementioned lenses is the same as the curve equation of the first embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the first embodiment, so an explanation in this regard will not be provided again.

(77) These parameters can be calculated from Table 3 and Table 4 as the following values, and the following conditions in the following table are satisfied.

(78) TABLE-US-00005 Embodiment 2 f[mm] 1.60 TL/(IMH*f)[mm.sup.−1] 3.67 EPD[mm] 0.71 FOV/CRA 5.30 Fno 2.20 f2/f3 1.30 FOV[°] 151.0 (R8/R9)/T45[mm.sup.−1] 11.25 f/f1 −0.48 R10/R11 6.00 R5/R6 2.16 (CT4 + CT6)/(CT5*f5)[mm.sup.−1] −0.89 f4/f5 −0.97 f5*f6/f[mm] −10.80 (R5*R6)/(f3*CT3) 7.64 tan(CRA)*BFL/IMH 0.38 f6/R11 3.62 CRA/(BFL*f)[°/mm.sup.2] 11.06 f4*R8/(R9*f5) −1.12 vd3/(nd3*f3*CT3)[mm.sup.−2] 2.01 (TL-BFL)/IMH 5.17

Third Embodiment

(79) Referring to FIGS. **3**A and **3**B, FIG. **3**A shows a schematic view of an optical lens assembly in accordance with a third embodiment of the present invention, FIG. **3**B shows, in order from left to right, the field curvature curve and the distortion curve of the third embodiment of the present invention, which illustrate the states of the field curvature curve and the distortion curve in both visible and infrared light bands. As shown in FIG. **3**A, the optical lens assembly includes, in order from an object side to an image side along an optical axis **390**: a first lens **310**, a second lens **320**, a stop **300**, a third lens **330**, a fourth lens **340**, a fifth lens **350**, a sixth lens **360**, an optical filter **370**, and an image plane **382**. The optical lens assembly can cooperate with an image sensor **381** disposed on an image plane **382**. The optical lens assembly has a total of six lenses with refractive power, but not is limited thereto.

(80) The first lens **310** with negative refractive power includes an object-side surface **311** and an image-side surface **312**, the object-side surface **311** of the first lens **310** is concave in a paraxial region thereof, the image-side surface **312** of the first lens **310** is concave in a paraxial region thereof, the object-side surface **311** and the image-side surface **312** of the first lens **310** are aspheric, and the first lens **310** is made of plastic.

(81) The second lens **320** with positive refractive power includes an object-side surface **321** and an image-side surface **322**, the object-side surface **321** of the second lens **320** is concave in a paraxial region thereof, the image-side surface **322** of the second lens **320** is convex in a paraxial region thereof, the object-side surface **321** and the image-side surface **322** of the second lens **320** are aspheric, and the second lens **320** is made of plastic.

(82) The third lens **330** with positive refractive power includes an object-side surface **331** and an image-side surface **332**, the object-side surface **331** of the third lens **330** is concave in a paraxial region thereof, the image-side surface **332** of the third lens **330** is convex in a paraxial region thereof, the object-side surface **331** and the image-side surface **332** of the third lens **330** are aspheric, and the third lens **330** is made of glass.

(83) The fourth lens **340** with positive refractive power includes an object-side surface **341** and an image-side surface **342**, the object-side surface **341** of the fourth lens **340** is convex in a paraxial region thereof, the image-side surface **342** of the fourth lens **340** is convex in a paraxial region thereof, the object-side surface **341** and the image-side surface **342** of the fourth lens **340** are

aspheric, and the fourth lens **340** is made of plastic.
(84) The fifth lens **350** with negative refractive power includes an object-side surface **351** and an image-side surface **352**, the object-side surface **351** of the fifth lens **350** is concave in a paraxial region thereof, the image-side surface **352** of the fifth lens **350** is concave in a paraxial region thereof, the object-side surface **351** and the image-side surface **352** of the fifth lens **350** are aspheric, and the fifth lens **350** is made of plastic.
(85) The sixth lens **360** with positive refractive power includes an object-side surface **361** and an image-side surface **362**, the object-side surface **361** of the sixth lens **360** is convex in a paraxial region thereof, the image-side surface **362** of the sixth lens **360** is concave in a paraxial region thereof, the object-side surface **361** and the image-side surface **362** of the sixth lens **360** are aspheric, and the sixth lens **360** is made of plastic.
(86) The optical filter **370** is made of glass, is located between the sixth lens **360** and the image plane **382**, and has no influence on the focal length of the optical lens assembly. In the present embodiment, the optical filter **370** is selected from filters that allow light in the visible light wavelengths, in the infrared light wavelengths or in both the visible and infrared light wavelengths to pass therethrough.
(87) The detailed optical data of the respective elements in the optical lens assembly of the third embodiment is shown in Table 5, and the aspheric coefficients of the lenses in the third embodiment is shown in Table 6.
(88) TABLE-US-00006 TABLE 5 Embodiment 3 f = 1.60 mm, Fno = 2.20, FOV = 151.0° Radius Refractive Abbe of Thickness/ index number Focal Surface curvature gap Material (nd) (vd) length 0 Object Infinity Infinity 1 First lens −20.062 (ASP) 1.195 plastic 1.544 56.0 −3.39 2 2.084 (ASP) 2.007 3 Second lens −32.391 (ASP) 3.165 plastic 1.661 20.4 20.23 4 −9.891 (ASP) 0.688 5 Stop Infinity 0.100 6 Third lens −5.910 (ASP) 0.906 glass 1.810 41.0 7.46 7 −3.201 (ASP) 0.040 8 Fourth lens 3.113 (ASP) 1.434 plastic 1.544 56.0 2.66 9 −2.284 (ASP) 0.076 10 Fifth lens −2.883 (ASP) 1.321 plastic 1.661 20.4 −2.01 11 2.954 (ASP) 0.049 12 Sixth lens 2.413 (ASP) 1.203 plastic 1.544 56.0 5.05 13 15.906 (ASP) 0.514 14 Optical filter Infinity 0.210 glass 1.517 64.2 15 Infinity 0.814 16 Image plane Infinity —
(89) TABLE-US-00007 TABLE 6 Aspheric Coefficients Surface 1 2 3 4 6 7 K: 8.4374E+00 −6.8631E−01 −9.3734E+01 5.0648E+00 −6.3532E−02 −4.0414E−03 A2: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A4: 4.0492E−03 2.0271E−03 −1.5465E−02 8.5315E−03 8.0487E−03 −4.7757E−03 A6: −1.8643E−04 5.8110E−04 2.0209E−03 3.4195E−03 −9.9287E−03 −5.9364E−03 A8: 6.0053E−06 5.5207E−04 −5.2698E−04 −2.4473E−03 0.0000E+00 −4.7113E−04 A10: −1.0271E−07 −1.2974E−04 −1.1615E−05 9.0971E−04 0.0000E+00 0.0000E+00 A12: 7.1657E−10 2.5967E−05 3.0248E−05 2.6311E−04 0.0000E+00 0.0000E+00 A14: 2.9179E−12 −2.7738E−06 −3.3262E−06 −2.2020E−04 0.0000E+00 0.0000E+00 A16: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A18: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A20: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 Surface 8 9 10 11 12 13 K: −2.3001E+00 7.8895E−02 −7.5700E−01 4.5709E−02 −1.9419E+00 −6.7920E+01 A2: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A4: −9.9627E−03 −8.3773E−03 −1.0946E−02 −1.6623E−02 −3.2110E−02 −4.3273E−03 A6: −5.2868E−04 3.4902E−03 −4.1931E−03 2.2114E−03 1.1141E−02 −6.5733E−04 A8: −2.7362E−03 5.0401E−04 2.9376E−03 6.7181E−04 −6.3251E−04 9.6498E−04 A10: −1.4172E−04 1.0706E−04 4.6696E−04 −2.6889E−04 −2.6798E−05 −1.1438E−04 A12: 2.8260E−04 −1.4807E−04 1.8662E−04 1.3064E−05 −2.3547E−05 5.7861E−06 A14: 3.8238E−05 9.0106E−05 −1.0814E−04 −3.7114E−06 4.1288E−06 5.1378E−06 A16: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A18: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A20: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00

(90) In the third embodiment, the curve equation of the aspheric surface profiles of the aforementioned lenses is the same as the curve equation of the first embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the first embodiment, so an explanation in this regard will not be provided again.
(91) These parameters can be calculated from Table 5 and Table 6 as the following values, and the following conditions in the following table are satisfied.
(92) TABLE-US-00008 Embodiment 3 f[mm] 1.60 TL/(IMH*f)[mm.sup.−1] 3.71 EPD[mm] 0.72 FOV/CRA 5.30 Fno 2.20 f2/f3 2.71 FOV[°] 151.0 (R8/R9)/T45[mm.sup.−1] 7.69 f/f1 −0.47 R10/R11 1.22 R5/R6 1.85 (CT4 + CT6)/(CT5*f5)[mm.sup.−1] −1.02 f4/f5 −1.33 f5*f6/f[mm] −6.33 (R5*R6)/(f3*CT3) 2.80 tan(CRA)*BFL/IMH 0.36 f6/R11 2.09 CRA/(BFL*f)[°/mm.sup.2] 11.57 f4*R8/(R9*f5) −1.05 vd3/(nd3*f3*CT3)[mm.sup.−2] 3.35 (TL-BFL)/IMH 5.28
Fourth Embodiment
(93) Referring to FIGS. **4**A and **4**B, FIG. **4**A shows a schematic view of an optical lens assembly in accordance with a fourth embodiment of the present invention, FIG. **4**B shows, in order from left to right, the field curvature curve and the distortion curve of the fourth embodiment of the present invention, which illustrate the states of the field curvature curve and the distortion curve in both visible and infrared light bands. As shown in FIG. **4**A, the optical lens assembly includes, in order from an object side to an image side along an optical axis **490**: a first lens **410**, a second lens **420**, a stop **400**, a third lens **430**, a fourth lens **440**, a fifth lens **450**, a sixth lens **460**, an optical filter **470**, and an image plane **482**. The optical lens assembly can cooperate with an image sensor **481** disposed on an image plane **482**. The optical lens assembly has a total of six lenses with refractive power, but not is limited thereto.
(94) The first lens **410** with negative refractive power includes an object-side surface **411** and an image-side surface **412**, the object-side surface **411** of the first lens **410** is concave in a paraxial region thereof, the image-side surface **412** of the first lens **410** is concave in a paraxial region thereof, the object-side surface **411** and the image-side surface **412** of the first lens **410** are aspheric, and the first lens **410** is made of plastic.
(95) The second lens **420** with positive refractive power includes an object-side surface **421** and an image-side surface **422**, the object-side surface **421** of the second lens **420** is concave in a paraxial region thereof, the image-side surface **422** of the second lens **420** is convex in a paraxial region thereof, the object-side surface **421** and the image-side surface **422** of the second lens **420** are aspheric, and the second lens **420** is made of plastic.
(96) The third lens **430** with positive refractive power includes an object-side surface **431** and an image-side surface **432**, the object-side surface **431** of the third lens **430** is concave in a paraxial region thereof, the image-side surface **432** of the third lens **430** is convex in a paraxial region thereof, the object-side surface **431** and the image-side surface **432** of the third lens **430** are aspheric, and the third lens **430** is made of glass.
(97) The fourth lens **440** with positive refractive power includes an object-side surface **441** and an image-side surface **442**, the object-side surface **441** of the fourth lens **440** is convex in a paraxial region thereof, the image-side surface **442** of the fourth lens **440** is convex in a paraxial region thereof, the object-side surface **441** and the image-side surface **442** of the fourth lens **440** are aspheric, and the fourth lens **440** is made of plastic.
(98) The fifth lens **450** with negative refractive power includes an object-side surface **451** and an image-side surface **452**, the object-side surface **451** of the fifth lens **450** is concave in a paraxial region thereof, the image-side surface **452** of the fifth lens **450** is concave in a paraxial region thereof, the object-side surface **451** and the image-side surface **452** of the fifth lens **450** are aspheric, and the fifth lens **450** is made of plastic.
(99) The sixth lens **460** with positive refractive power includes an object-side surface **461** and an image-side surface **462**, the object-side surface **461** of the sixth lens **460** is convex in a paraxial region thereof, the image-side surface **462** of the sixth lens **460** is convex in a paraxial region

thereof, the object-side surface **461** and the image-side surface **462** of the sixth lens **460** are aspheric, and the sixth lens **460** is made of plastic.
(100) The optical filter **470** is made of glass, is located between the sixth lens **460** and the image plane **482**, and has no influence on the focal length of the optical lens assembly. In the present embodiment, the optical filter **470** is selected from filters that allow light in the visible light wavelengths, in the infrared light wavelengths or in both the visible and infrared light wavelengths to pass therethrough.
(101) The detailed optical data of the respective elements in the optical lens assembly of the fourth embodiment is shown in Table 7, and the aspheric coefficients of the lenses in the fourth embodiment is shown in Table 8.
(102) TABLE-US-00009 TABLE 7 Embodiment 4 f = 1.63 mm, Fno = 2.20, FOV = 151.0° Radius Refractive Abbe of Thickness/ index number Focal Surface curvature gap Material (nd) (vd) length 0 Object Infinity Infinity 1 First lens −11.185 (ASP) 1.689 plastic 1.544 56.0 −3.21 2 2.186 (ASP) 1.507 3 Second lens −15.275 (ASP) 2.929 plastic 1.661 20.4 20.38 4 −7.742 (ASP) 0.439 5 Stop Infinity 0.218 6 Third lens −4.100 0.909 glass 1.773 49.6 6.44 7 −2.469 0.093 8 Fourth lens 3.443 (ASP) 1.635 plastic 1.544 56.0 2.97 9 −2.548 (ASP) 0.097 10 Fifth lens −3.197 (ASP) 0.528 plastic 1.661 20.4 −2.24 11 2.982 (ASP) 0.100 12 Sixth lens 3.144 (ASP) 1.595 plastic 1.544 56.0 4.60 13 −10.264 (ASP) 0.334 14 Optical filter Infinity 0.210 glass 1.517 64.2 15 Infinity 1.240 16 Image plane Infinity —
(103) TABLE-US-00010 TABLE 8 Aspheric Coefficients Surface 1 2 3 4 6 7 K: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A2: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A4: 4.9727E−03 −1.1483E−03 −1.3516E−02 1.1132E−02 0.0000E+00 0.0000E+00 A6: −2.2374E−04 −1.1980E−03 2.5645E−04 6.1876E−03 0.0000E+00 0.0000E+00 A8: 6.4865E−06 1.3153E−03 6.3664E−05 1.9481E−03 0.0000E+00 0.0000E+00 A10: −9.0668E−08 −2.5137E−04 2.0044E−05 −5.9241E−03 0.0000E+00 0.0000E+00 A12: 1.7640E−10 −1.8944E−06 −3.3125E−06 5.7376E−03 0.0000E+00 0.0000E+00 A14: 6.8234E−12 −7.0882E−07 3.5748E−07 −1.3848E−03 0.0000E+00 0.0000E+00 A16: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A18: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A20: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 Surface 8 9 10 11 12 13 K: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A2: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A4: −2.5656E−03 −3.6569E−03 −1.5895E−02 −1.8696E−02 −1.4572E−02 6.5642E−03 A6: 1.3030E−03 8.0258E−04 −1.5669E−03 3.0348E−03 5.5254E−03 3.5580E−03 A8: −5.2569E−04 2.9238E−04 1.2168E−03 3.4769E−04 −8.2186E−04 −3.3664E−04 A10: 8.8360E−05 −2.4644E−05 −3.7971E−04 −1.9837E−04 5.1995E−05 −1.1634E−04 A12: −1.6561E−05 −1.5776E−05 1.1025E−05 −1.6772E−06 1.7466E−06 1.5802E−05 A14: −1.0467E−05 5.8510E−07 2.0975E−05 9.3389E−07 3.8029E−08 5.5785E−06 A16: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A18: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A20: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00
(104) In the fourth embodiment, the curve equation of the aspheric surface profiles of the aforementioned lenses is the same as the curve equation of the first embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the first embodiment, so an explanation in this regard will not be provided again.
(105) These parameters can be calculated from Table 7 and Table 8 as the following values, and the following conditions in the following table are satisfied.
(106) TABLE-US-00011 Embodiment 4 f[mm] 1.63 TL/(IMH*f)[mm.sup.−1] 3.61 EPD[mm] 0.73 FOV/CRA 6.31 Fno 2.20 f2/f3 3.17 FOV[°] 151.0 (R8/R9)/T45[mm.sup.−1] 8.20 f/f1 −0.51 R10/R11 0.95 R5/R6 1.66 (CT4 + CT6)/(CT5*f5)[mm.sup.−1] −2.73 f4/f5 −1.33 f5*f6/f[mm]

−6.34 (R5*R6)/(f3*CT3) 1.73 tan(CRA)*BFL/IMH 0.34 f6/R11 1.46 CRA/(BFL*f)[°/mm.sup.2] 8.24 f4*R8/(R9*f5) −1.06 vd3/(nd3*f3*CT3)[mm.sup.−2] 4.78 (TL-BFL)/IMH 5.09

Fifth Embodiment

(107) Referring to FIGS. **5**A and **5**B, FIG. **5**A shows a schematic view of an optical lens assembly in accordance with a fifth embodiment of the present invention, FIG. **5**B shows, in order from left to right, the field curvature curve and the distortion curve of the fifth embodiment of the present invention, which illustrate the states of the field curvature curve and the distortion curve in both visible and infrared light bands. As shown in FIG. **5**A, the optical lens assembly includes, in order from an object side to an image side along an optical axis **590**: a first lens **510**, a second lens **520**, a stop **500**, a third lens **530**, a fourth lens **540**, a fifth lens **550**, a sixth lens **560**, an optical filter **570**, and an image plane **582**. The optical lens assembly can cooperate with an image sensor **581** disposed on an image plane **582**. The optical lens assembly has a total of six lenses with refractive power, but not is limited thereto.

(108) The first lens **510** with negative refractive power includes an object-side surface **511** and an image-side surface **512**, the object-side surface **511** of the first lens **510** is concave in a paraxial region thereof, the image-side surface **512** of the first lens **510** is concave in a paraxial region thereof, the object-side surface **511** and the image-side surface **512** of the first lens **510** are aspheric, and the first lens **510** is made of plastic.

(109) The second lens **520** with positive refractive power includes an object-side surface **521** and an image-side surface **522**, the object-side surface **521** of the second lens **520** is concave in a paraxial region thereof, the image-side surface **522** of the second lens **520** is convex in a paraxial region thereof, the object-side surface **521** and the image-side surface **522** of the second lens **520** are aspheric, and the second lens **520** is made of plastic.

(110) The third lens **530** with positive refractive power includes an object-side surface **531** and an image-side surface **532**, the object-side surface **531** of the third lens **530** is concave in a paraxial region thereof, the image-side surface **532** of the third lens **530** is convex in a paraxial region thereof, the object-side surface **531** and the image-side surface **532** of the third lens **530** are aspheric, and the third lens **530** is made of glass.

(111) The fourth lens **540** with positive refractive power includes an object-side surface **541** and an image-side surface **542**, the object-side surface **541** of the fourth lens **540** is convex in a paraxial region thereof, the image-side surface **542** of the fourth lens **540** is convex in a paraxial region thereof, the object-side surface **541** and the image-side surface **542** of the fourth lens **540** are aspheric, and the fourth lens **540** is made of plastic.

(112) The fifth lens **550** with negative refractive power includes an object-side surface **551** and an image-side surface **552**, the object-side surface **551** of the fifth lens **550** is concave in a paraxial region thereof, the image-side surface **552** of the fifth lens **550** is concave in a paraxial region thereof, the object-side surface **551** and the image-side surface **552** of the fifth lens **550** are aspheric, and the fifth lens **550** is made of plastic.

(113) The sixth lens **560** with positive refractive power includes an object-side surface **561** and an image-side surface **562**, the object-side surface **561** of the sixth lens **560** is convex in a paraxial region thereof, the image-side surface **562** of the sixth lens **560** is convex in a paraxial region thereof, the object-side surface **561** and the image-side surface **562** of the sixth lens **560** are aspheric, and the sixth lens **560** is made of plastic.

(114) The optical filter **570** is made of glass, is located between the sixth lens **560** and the image plane **582**, and has no influence on the focal length of the optical lens assembly. In the present embodiment, the optical filter **570** is selected from filters that allow light in the visible light wavelengths, in the infrared light wavelengths or in both the visible and infrared light wavelengths to pass therethrough.

(115) The cover glass **583** is made of glass, is disposed between the optical filter **570** and the image plane **582**, and has no influence on the focal length of the optical lens assembly.

(116) The detailed optical data of the respective elements in the optical lens assembly of the fifth embodiment is shown in Table 9, and the aspheric coefficients of the lenses in the fifth embodiment is shown in Table 10.

(117) TABLE-US-00012 TABLE 9 Embodiment 5 f = 3.21 mm, Fno = 1.60, FOV = 142.4° Radius Refractive Abbe of Thickness/ index number Focal Surface curvature gap Material (nd) (vd) length 0 Object Infinity Infinity 1 First lens −407.803 (ASP) 1.080 plastic 1.544 56.0 −5.65 2 3.112 (ASP) 2.233 3 Second lens −15.396 (ASP) 5.412 plastic 1.661 20.4 41.88 4 −11.323 (ASP) 1.293 5 Stop Infinity 0.245 6 Third lens −17.702 1.051 glass 1.804 46.6 10.70 7 −5.959 0.091 8 Fourth lens 9.246 (ASP) 1.596 plastic 1.544 56.0 9.50 9 −11.111 (ASP) 0.276 10 Fifth lens −14.541 (ASP) 0.588 plastic 1.661 20.4 −5.08 11 4.491 (ASP) 0.347 12 Sixth lens 5.390 (ASP) 1.920 plastic 1.544 56.0 6.03 13 −7.400 (ASP) 2.470 14 Optical filter Infinity 0.300 glass 1.517 64.2 15 Infinity 2.693 16 Cover glass Infinity 0.400 glass 1.517 64.2 17 Infinity 0.045 18 Image plane Infinity —

(118) TABLE-US-00013 TABLE 10 Aspheric Coefficients Surface 1 2 3 4 6 7 K 0.0000E+00 0.0000E+00 1.6519E+01 2.9262E+00 0.0000E+00 0.0000E+00 A2: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A4: 4.3544E−04 −9.2047E−04 −2.3810E−03 −1.9569E−05 0.0000E+00 0.0000E+00 A6: −1.9076E−06 −1.1352E−04 6.0713E−06 1.6928E−04 0.0000E+00 0.0000E+00 A8: −5.9526E−08 1.0546E−05 −2.3650E−06 −4.9663E−06 0.0000E+00 0.0000E+00 A10: 1.5614E−09 −1.5084E−06 6.1320E−07 2.3963E−07 0.0000E+00 0.0000E+00 A12: −3.8975E−13 4.0731E−08 −3.3879E−08 8.9991E−08 0.0000E+00 0.0000E+00 A14: 1.0154E−13 −2.8417E−09 −6.7117E−10 −6.7792E−09 0.0000E+00 0.0000E+00 A16: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A18: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A20: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 Surface 8 9 10 11 12 13 K: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 −5.9049E−02 3.6509E−01 A2: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A4: −3.8089E−03 −4.8614E−03 −2.6389E−03 −6.7623E−03 −5.1373E−03 4.9444E−04 A6: 4.1005E−05 1.2342E−04 −1.2709E−04 1.9948E−04 2.7971E−04 −6.2153E−05 A8: 2.4196E−06 4.3950E−06 2.0894E−05 −5.0628E−06 −4.9268E−06 8.8055E−06 A10: −7.2068E−07 −1.0506E−06 −1.2821E−06 −8.6227E−07 −5.0141E−07 −1.0789E−07 A12: −1.7121E−08 −3.1361E−08 −3.4092E−08 −2.7022E−08 −1.0925E−08 2.5324E−10 A14: 2.5238E−10 −1.7309E−09 −7.1154E−09 −3.0898E−10 −1.4492E−09 −1.4938E−09 A16: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A18: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A20: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00

(119) In the fifth embodiment, the curve equation of the aspheric surface profiles of the aforementioned lenses is the same as the curve equation of the first embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the first embodiment, so an explanation in this regard will not be provided again.

(120) These parameters can be calculated from Table 9 and Table 10 as the following values, and the following conditions in the following table are satisfied.

(121) TABLE-US-00014 Embodiment 5 f[mm] 3.21 TL/(IMH*f)[mm.sup.−1] 1.99 EPD[mm] 1.97 FOV/CRA 11.10 Fno 1.60 f2/f3 3.91 FOV[°] 142.4 (R8/R9)/T45[mm.sup.−1] 2.76 f/f1 −0.57 R10/R11 0.83 R5/R6 2.97 (CT4 + CT6)/(CT5*f5)[mm.sup.−1] −1.18 f4/f5 −1.87 f5*f6/f[mm] −9.53 (R5*R6)/(f3*CT3) 9.38 tan(CRA)*BFL/IMH 0.39 f6/R11 1.12 CRA/(BFL*f)[°/mm.sup.2] 0.68 f4*R8/(R9*f5) −1.43 vd3/(nd3*f3*CT3)[mm.sup.−2] 2.30 (TL-BFL)/IMH 4.68

Sixth Embodiment

(122) Referring to FIGS. **6**A and **6**B, FIG. **6**A shows a schematic view of an optical lens assembly in accordance with a sixth embodiment of the present invention, FIG. **6**B shows, in order from left to right, the field curvature curve and the distortion curve of the sixth embodiment of the present invention, which illustrate the states of the field curvature curve and the distortion curve in both

visible and infrared light bands. As shown in FIG. **6**A, the optical lens assembly includes, in order from an object side to an image side along an optical axis **690**: a first lens **610**, a second lens **620**, a stop **600**, a third lens **630**, a fourth lens **640**, a fifth lens **650**, a sixth lens **660**, an optical filter **670**, and an image plane **682**. The optical lens assembly can cooperate with an image sensor **681** disposed on an image plane **682**. The optical lens assembly has a total of six lenses with refractive power, but not is limited thereto.

(123) The first lens **610** with negative refractive power includes an object-side surface **611** and an image-side surface **612**, the object-side surface **611** of the first lens **610** is concave in a paraxial region thereof, the image-side surface **612** of the first lens **610** is concave in a paraxial region thereof, the object-side surface **611** and the image-side surface **612** of the first lens **610** are aspheric, and the first lens **610** is made of plastic.

(124) The second lens **620** with positive refractive power includes an object-side surface **621** and an image-side surface **622**, the object-side surface **621** of the second lens **620** is concave in a paraxial region thereof, the image-side surface **622** of the second lens **620** is convex in a paraxial region thereof, the object-side surface **621** and the image-side surface **622** of the second lens **620** are aspheric, and the second lens **620** is made of plastic.

(125) The third lens **630** with positive refractive power includes an object-side surface **631** and an image-side surface **632**, the object-side surface **631** of the third lens **630** is concave in a paraxial region thereof, the image-side surface **632** of the third lens **630** is convex in a paraxial region thereof, the object-side surface **631** and the image-side surface **632** of the third lens **630** are aspheric, and the third lens **630** is made of glass.

(126) The fourth lens **640** with positive refractive power includes an object-side surface **641** and an image-side surface **642**, the object-side surface **641** of the fourth lens **640** is convex in a paraxial region thereof, the image-side surface **642** of the fourth lens **640** is convex in a paraxial region thereof, the object-side surface **641** and the image-side surface **642** of the fourth lens **640** are aspheric, and the fourth lens **640** is made of plastic.

(127) The fifth lens **650** with negative refractive power includes an object-side surface **651** and an image-side surface **652**, the object-side surface **651** of the fifth lens **650** is concave in a paraxial region thereof, the image-side surface **652** of the fifth lens **650** is concave in a paraxial region thereof, the object-side surface **651** and the image-side surface **652** of the fifth lens **650** are aspheric, and the fifth lens **650** is made of plastic.

(128) The sixth lens **660** with positive refractive power includes an object-side surface **661** and an image-side surface **662**, the object-side surface **661** of the sixth lens **660** is convex in a paraxial region thereof, the image-side surface **662** of the sixth lens **660** is convex in a paraxial region thereof, the object-side surface **661** and the image-side surface **662** of the sixth lens **660** are aspheric, and the sixth lens **660** is made of plastic.

(129) The optical filter **670** is made of glass, is located between the sixth lens **660** and the image plane **682**, and has no influence on the focal length of the optical lens assembly. In the present embodiment, the optical filter **670** is selected from filters that allow light in the visible light wavelengths, in the infrared light wavelengths or in both the visible and infrared light wavelengths to pass therethrough.

(130) The detailed optical data of the respective elements in the optical lens assembly of the sixth embodiment is shown in Table 11, and the aspheric coefficients of the lenses in the sixth embodiment is shown in Table 12.

(131) TABLE-US-00015 TABLE 11 Embodiment 6 f = 1.31 mm, Fno = 2.20, FOV = 152.0° Radius Refractive Abbe of Thickness/ index number Focal Surface curvature gap Material (nd) (vd) length 0 Object Infinity Infinity 1 First lens −25.078 (ASP) 1.641 plastic 1.544 56.0 −3.05 2 1.826 (ASP) 2.834 3 Second lens −37.026 (ASP) 2.561 plastic 1.643 22.5 13.42 4 −7.228 (ASP) 0.521 5 Third lens −5.811 (ASP) 1.595 glass 1.803 45.5 14.52 6 −4.360 (ASP) 0.295 7 Stop Infinity −0.113 8 fourth lens 2.544 (ASP) 1.395 plastic 1.544 56.0 2.37 9 −2.132 (ASP) 0.166 10

Fifth lens −2.980 (ASP) 0.766 plastic 1.671 19.2 −2.06 11 2.899 (ASP) 0.230 12 Sixth lens 3.300 (ASP) 1.251 plastic 1.544 56.0 4.51 13 −8.401 (ASP) 0.814 14 Optical filter Infinity 0.210 glass 1.517 64.2 15 Infinity 0.500 16 Image plane Infinity —

(132) TABLE-US-00016 TABLE 12 Aspheric Coefficients Surface 1 2 3 4 5 6 K 8.7265E+00 −6.5690E−01 8.1274E+01 1.3481E+01 −3.4551E−02 1.3036E−02 A2: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A4: 3.3043E−03 −4.6951E−03 −1.6510E−02 3.6290E−03 1.5172E−04 −8.6829E−05 A6: −1.5083E−04 −7.2726E−04 2.7779E−03 6.8950E−03 −1.2584E−04 −3.6687E−05 A8: 4.0783E−06 1.1870E−03 −1.2065E−03 −7.0549E−03 −1.1078E−05 1.8783E−04 A10: −6.0233E−08 −3.0711E−04 2.6837E−04 5.6473E−03 −1.1078E−05 1.5516E−04 A12: 4.5442E−10 2.9173E−05 −2.1659E−05 −2.0921E−03 0.0000E+00 0.0000E+00 A14: −1.2141E−12 −1.1866E−06 4.6477E−07 3.2627E−04 0.0000E+00 0.0000E+00 A16: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A18: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A20: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 Surface 8 9 10 11 12 13 K −4.9745E−01 −8.2480E−02 1.0036E+00 3.0383E−01 −4.5241E−01 −8.8599E+01 A2: 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 0.0000E+00 A4: 3.7565E−04 −3.4187E−02 −9.6252E−02 −8.9762E−02 −1.6867E−02 4.2641E−02 A6: 1.1823E−02 1.5851E−01 1.9695E−01 1.1512E−01 9.1377E−03 −1.8697E−02 A8: −6.5492E−02 −3.3350E−01 −1.4931E−01 −1.1349E−01 −1.2353E−02 9.9061E−03 A10: 5.5853E−02 7.8140E−01 −6.3514E−02 1.6139E−01 2.4328E−02 −1.4272E−02 A12: 2.9571E−01 −1.6450E+00 1.0783E−01 −2.3110E−01 −2.4257E−02 1.2828E−02 A14: −8.7564E−01 2.1664E+00 2.9767E−02 2.0869E−01 1.3282E−02 −6.1114E−03 A16: 9.7409E−01 −1.6510E+00 −1.1506E−01 −1.0873E−01 −4.2075E−03 1.6268E−03 A18: −4.9353E−01 6.7494E−01 7.9070E−02 3.0410E−02 7.2826E−04 −2.3337E−04 A20: 9.4108E−02 −1.1450E−01 −1.8469E−02 −3.5473E−03 −5.3355E−05 1.4189E−05

(133) In the sixth embodiment, the curve equation of the aspheric surface profiles of the aforementioned lenses is the same as the curve equation of the first embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the first embodiment, so an explanation in this regard will not be provided again.

(134) These parameters can be calculated from Table 11 and Table 12 as the following values, and the following conditions in the following table are satisfied.

(135) TABLE-US-00017 Embodiment 6 f[mm] 1.31 TL/(IMH*f)[mm.sup.−1] 4.86 EPD[mm] 0.58 FOV/CRA 5.28 Fno 2.20 f2/f3 0.92 FOV[°] 152.0 (R8/R9)/T45[mm.sup.−1] 4.30 f/f1 −0.43 R10/R11 0.88 R5/R6 1.33 (CT4 + CT6)/(CT5*f5)[mm.sup.−1] −1.68 f4/f5 −1.15 f5*f6/f[mm] −7.11 (R5*R6)/(f3*CT3) 1.09 tan(CRA)*BFL/IMH 0.36 f6/R11 1.37 CRA/(BFL*f)[°/mm.sup.2] 14.44 f4*R8/(R9*f5) −0.82 vd3/(nd3*f3*CT3)[mm.sup.−2] 1.09 (TL-BFL)/IMH 5.70

Seventh Embodiment

(136) Referring to FIG. **7**, which shows a schematic view of a photographing module in accordance with a seventh embodiment of the present invention. The photographing module **10** includes a lens barrel **11**, an optical lens assembly **12** and an image sensor **13**. The optical lens assembly **12** is the optical lens assembly of any one of the above embodiments and is disposed in the lens barrel **11**. The image sensor **13** is disposed on an image plane **121** of the optical lens assembly **12** and is an electronic image sensor (such as, CMOS or CCD) with good photosensitivity and low noise to really present the imaging quality of the optical lens assembly.

(137) For the optical lens assembly in the present invention, the lenses can be made of plastic or glass. If the lens is made of plastic, it is conducive to reducing the manufacturing cost. If the lens is made of glass, it is conducive to enhancing the degree of freedom in the arrangement of refractive power of the optical lens assembly. Moreover, any of the object-side and image-side surfaces of a respective lens of the optical lens assembly can be aspheric, and the aspheric surface can have any profile shape other than the profile shape of a spherical surface, so more variables can be used in

the design of aspheric surfaces (than spherical surfaces), which is conducive to reducing the aberration and the number of lenses, as well as the total length of the optical lens assembly.
(138) In the optical lens assembly of the present invention, the optical filter is made of, but not limited to, glass and can be made of other materials with high Abbe numbers.
(139) For the optical lens assembly in the present invention, if the surface shape of a respective lens surface of a respective lens with refractive power is convex and the location of the convex portion of the respective lens surface of the respective lens is not defined, the convex portion is typically located in a paraxial region of the respective lens surface of the respective lens. If the surface shape of a respective lens surface of a respective lens is concave and the location of the concave portion of the respective lens surface of the respective lens is not defined, the concave portion is typically located in a paraxial region of the respective lens surface of the respective lens.
(140) The optical lens assembly of the present invention can be used in focus-adjustable optical systems according to the actual requirements and have good aberration correction ability and better image quality. The optical lens assembly of the present invention can also be used in electronic imaging systems, such as, 3D image capturing device, wearable display of virtual reality (VR) or augmented reality (AR), game player, surveillance camera, digital camera, mobile device, tablet computer, household electronic device or vehicle camera.

Claims

1. An optical lens assembly comprising, in order from an object side to an image side: a first lens with negative refractive power, comprising an object-side surface and an image-side surface, the object-side surface of the first lens being concave in a paraxial region thereof, and the image-side surface of the first lens being concave in a paraxial region thereof; a second lens with positive refractive power, comprising an object-side surface and an image-side surface, the object-side surface of the second lens being concave in a paraxial region thereof, and the image-side surface of the second lens being convex in a paraxial region thereof; a third lens with positive refractive power, comprising an object-side surface and an image-side surface, the object-side surface of the third lens being concave in a paraxial region thereof, and the image-side surface of the third lens being convex in a paraxial region thereof; a fourth lens with positive refractive power, comprising an object-side surface and an image-side surface, the object-side surface of the fourth lens being convex in a paraxial region thereof, and the image-side surface of the fourth lens being convex in a paraxial region thereof; a fifth lens with negative refractive power, comprising an object-side surface and an image-side surface, the object-side surface of the fifth lens being concave in a paraxial region thereof, and the image-side surface of the fifth lens being concave in a paraxial region thereof; and a sixth lens with positive refractive power, comprising an object-side surface and an image-side surface, and the object-side surface of the sixth lens being convex in a paraxial region thereof; wherein a chief ray angle incident on an image plane at a maximum view angle of the optical lens assembly is CRA, a distance from the image-side surface of the sixth lens to the image plane along an optical axis is BFL, a focal length of the optical lens assembly is f, and the following condition is satisfied: $0.54° / mm^2 < CRA / (BFL * f) < 17.33° / mm^2$.
2. The optical lens assembly as claimed in claim 1, wherein a radius of curvature of the object-side surface of the third lens is R5, a radius of curvature of the image-side surface of the third lens is R6, and the following condition is satisfied: $1.07 < R5 / R6 < 3.56$.
3. The optical lens assembly as claimed in claim 1, wherein a radius of curvature of the object-side surface of the third lens is R5, a radius of curvature of the image-side surface of the third lens is R6, a focal length of the third lens is f3, a thickness of the third lens along the optical axis is CT3, and the following condition is satisfied: $0.88 < (R5 * R6) / (f3 * CT3) < 11.25$.
4. The optical lens assembly as claimed in claim 1, wherein a focal length of the sixth lens is f6, a

radius of curvature of the object-side surface of the sixth lens is R11, and the following condition is satisfied: $0.89 < f6 / R11 < 4.35$.
5. The optical lens assembly as claimed in claim 1, wherein a distance from the object-side surface of the first lens to the image plane along the optical axis is TL, the distance from the image-side surface of the sixth lens to the image plane along the optical axis is BFL, a maximum image height of the optical lens assembly is IMH, and the following condition is satisfied:
$3.74 < (TL - BFL) / IMH < 6.84$.
6. The optical lens assembly as claimed in claim 1, wherein a maximum field of view of the optical lens assembly is FOV, the chief ray angle incident on the image plane at the maximum view angle of the optical lens assembly is CRA, and the following condition is satisfied:
$4.23 < FOV / CRA < 13.32$.
7. The optical lens assembly as claimed in claim 1, wherein a radius of curvature of the image-side surface of the fifth lens is R10, a radius of curvature of the object-side surface of the sixth lens is R11, and the following condition is satisfied: $0.67 < R10 / R11 < 7.2$.
8. The optical lens assembly as claimed in claim 1, wherein the chief ray angle incident on the image plane at the maximum view angle of the optical lens assembly is CRA, the distance from the image-side surface of the sixth lens to the image plane along the optical axis is BFL, a maximum image height of the optical lens assembly is IMH, and the following condition is satisfied:
$0.27 < \tan(CRA) * BFL / IMH < 0.47$.
9. The optical lens assembly as claimed in claim 1, wherein an Abbe number of the third lens is vd3, a refractive index of the third lens is nd3, a focal length of the third lens is f3, a thickness of the third lens along the optical axis is CT3, and the following condition is satisfied:
$0.87mm^{-2} < vd3 / (nd3 * f3 * CT3) < 5.74mm^{-2}$.
10. An optical lens assembly comprising, in order from an object side to an image side: a first lens with negative refractive power, comprising an object-side surface and an image-side surface, the object-side surface of the first lens being concave in a paraxial region thereof, and the image-side surface of the first lens being concave in a paraxial region thereof; a second lens with positive refractive power, comprising an object-side surface and an image-side surface, the object-side surface of the second lens being concave in a paraxial region thereof, and the image-side surface of the second lens being convex in a paraxial region thereof; a third lens with positive refractive power, comprising an object-side surface and an image-side surface, the object-side surface of the third lens being concave in a paraxial region thereof, the image-side surface of the third lens being convex in a paraxial region thereof, and the third lens being made of glass; a fourth lens with positive refractive power, comprising an object-side surface and an image-side surface, the object-side surface of the fourth lens being convex in a paraxial region thereof, and the image-side surface of the fourth lens being convex in a paraxial region thereof; a fifth lens with negative refractive power, comprising an object-side surface and an image-side surface, the object-side surface of the fifth lens being concave in a paraxial region thereof, and the image-side surface of the fifth lens being concave in a paraxial region thereof; and a sixth lens with positive refractive power, comprising an object-side surface and an image-side surface, the object-side surface of the sixth lens being convex in a paraxial region thereof, and the image-side surface of the sixth lens being concave in a paraxial region thereof; wherein the optical lens assembly comprising a stop disposed between the second lens and the fourth lens, a chief ray angle incident on an image plane at a maximum view angle of the optical lens assembly is CRA, a distance from the image-side surface of the sixth lens to the image plane along an optical axis is BFL, a focal length of the optical lens assembly is f, and the following condition is satisfied:
$8.85° / mm^2 < CRA / (BFL * f) < 13.88° / mm^2$.
11. The optical lens assembly as claimed in claim 10, wherein a radius of curvature of the object-side surface of the third lens is R5, a radius of curvature of the image-side surface of the third lens is R6, a focal length of the third lens is f3, a thickness of the third lens along the optical axis is

CT3, and the following condition is satisfied: $2.24 < (R5 * R6) / (f3 * CT3) < 9.17$.
12. The optical lens assembly as claimed in claim 10, wherein a focal length of the sixth lens is f6, a radius of curvature of the object-side surface of the sixth lens is R11, and the following condition is satisfied: $1.68 < f6 / R11 < 4.35$.
13. The optical lens assembly as claimed in claim 10, wherein a maximum field of view of the optical lens assembly is FOV, the chief ray angle incident on the image plane at the maximum view angle of the optical lens assembly is CRA, and the following condition is satisfied:
$4.23 < FOV / CRA < 6.36$.
14. The optical lens assembly as claimed in claim 10, wherein an Abbe number of the third lens is vd3, a refractive index of the third lens is nd3, a focal length of the third lens is f3, a thickness of the third lens along the optical axis is CT3, and the following condition is satisfied:
$1.61mm^{-2} < vd3 / (nd3 * f3 * CT3) < 4.02mm^{-2}$.
15. A photographing module, comprising: a lens barrel; an optical lens assembly disposed in the lens barrel; and an image sensor disposed on an image plane of the optical lens assembly; wherein the optical lens assembly comprising, in order from an object side to an image side: a first lens with negative refractive power, comprising an object-side surface and an image-side surface, the object-side surface of the first lens being concave in a paraxial region thereof, and the image-side surface of the first lens being concave in a paraxial region thereof; a second lens with positive refractive power, comprising an object-side surface and an image-side surface, the object-side surface of the second lens being concave in a paraxial region thereof, and the image-side surface of the second lens being convex in a paraxial region thereof; a third lens with positive refractive power, comprising an object-side surface and an image-side surface, the object-side surface of the third lens being concave in a paraxial region thereof, and the image-side surface of the third lens being convex in a paraxial region thereof; a fourth lens with positive refractive power, comprising an object-side surface and an image-side surface, the object-side surface of the fourth lens being convex in a paraxial region thereof, and the image-side surface of the fourth lens being convex in a paraxial region thereof; a fifth lens with negative refractive power, comprising an object-side surface and an image-side surface, the object-side surface of the fifth lens being concave in a paraxial region thereof, and the image-side surface of the fifth lens being concave in a paraxial region thereof; and a sixth lens with positive refractive power, comprising an object-side surface and an image-side surface, and the object-side surface of the sixth lens being convex in a paraxial region thereof; wherein a chief ray angle incident on an image plane at a maximum view angle of the optical lens assembly is CRA, a distance from the image-side surface of the sixth lens to the image plane along an optical axis is BFL, a focal length of the optical lens assembly is f, and the following condition is satisfied: $0.54° / mm^2 < CRA / (BFL * f) < 17.33° / mm^2$.
16. The photographing module as claimed in claim 15, wherein a radius of curvature of the object-side surface of the third lens is R5, a radius of curvature of the image-side surface of the third lens is R6, a focal length of the third lens is f3, a thickness of the third lens along the optical axis is CT3, and the following condition is satisfied: $0.88 < (R5 * R6) / (f3 * CT3) < 11.25$.
17. The photographing module as claimed in claim 15, wherein a focal length of the sixth lens is f6, a radius of curvature of the object-side surface of the sixth lens is R11, and the following condition is satisfied: $0.89 < f6 / R11 < 4.35$.
18. The photographing module as claimed in claim 15, wherein a maximum field of view of the optical lens assembly is FOV, the chief ray angle incident on the image plane at the maximum view angle of the optical lens assembly is CRA, and the following condition is satisfied:
$4.23 < FOV / CRA < 13.32$.
19. The photographing module as claimed in claim 15, wherein a radius of curvature of the image-side surface of the fifth lens is R10, a radius of curvature of the object-side surface of the sixth lens is R11, and the following condition is satisfied: $0.67 < R10 / R11 < 7.2$.
20. The photographing module as claimed in claim 15, wherein an Abbe number of the third lens is

vd3, a refractive index of the third lens is nd3, a focal length of the third lens is f3, a thickness of the third lens along the optical axis is CT3, and the following condition is satisfied:
$0.87\text{mm}^{-2} < vd3 / (nd3 * f3 * CT3) < 5.74\text{mm}^{-2}$.